

COURSE SYLLABUS

CSC13008 – Web Application Development

1. GENERAL INFORMATION

Course name:	Web Application Development
Course name (in Vietnamese):	Phát triển ứng dụng web
Course ID:	CSC13008
Knowledge block:	Software Engineering
Number of credits:	4
Credit hours for theory:	45
Credit hours for practice:	30
Credit hours for self-study:	90
Prerequisite:	Fundamentals of Programming, Database
Prior-course:	None
Instructors:	

2. COURSE DESCRIPTION

This course aims to provide students with an overview of web application development, focusing on both Front-end and Back-end aspects. Key concepts include web design principles, client-server architecture, dynamic web development using HTML5, CSS3, JavaScript, NodeJS, Express, and databases (SQL/NoSQL). Additionally, it introduces updated topics on SSR (Server-Side Rendering) and CSR (Client-Side Rendering) architectures, front-end development with ReactJS using state management and API integration, as well as modern CSS frameworks such as Tailwind or Bootstrap. Students will practice building real-world web applications, enhancing teamwork and project presentation skills.

3. COURSE GOALS

At the end of the course, students are able to

ID	Description	Program LOs
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G1	Work in teams (2–3 people) to develop real-world web applications of small to medium scale.	
G2	Self-study, read and comprehend specialized English documents, and present topics related to web development.	
G3	Master web application architecture and Internet technologies, including SSR and CSR.	
G4	Apply a web application development process to develop a project.	
G5	Design and build a static website using HTML5, CSS3, and frameworks such as Tailwind or Bootstrap.	
G6	English translation: Design and build a dynamic web application with a backend API using NodeJS, Express, and database connectivity.	
G7	Understand and apply modern frontend with ReactJS, especially state management and API integration.	

4. COURSE OUTCOMES

CO	Description	I/T/U
G1.1	Work in a small-to-medium web application development team.	ITU
G1.2	English translation: Explain specialized IT English terminology.	I
G2.2	Prepare posters and presentation slides.	U
G3.1	Explain the basic concepts of the Web and the Internet.	I
G3.2	Describe client-side/server-side architecture, SSR/CSR.	IT
G3.3	Explain the principles of web design.	IT
G3.4	Apply interface design principles.	U
G4.1	Use web development technologies and tools.	TU
G4.2	Plan, implement, debug, and deploy web applications.	TU
G5.1	Explain current HTML, CSS standards and CSS frameworks.	IT
G5.2	Explain DOM, JavaScript behavior, and SSR/CSR paradigms.	IT

G5.3	Apply basic programming in HTML, CSS, and JavaScript.	TU
G5.4	Responsive website design.	TU
G5.5	Website authentication according to W3C standards.	TU
G6.1	Understand JavaScript and NodeJS.	IT
G6.2	Organize NodeJS project by modules.	TU
G6.3	Build a web server with NodeJS.	TU
G6.4	Build a web app, RESTful API with Express.	TU
G6.5	Connect to SQL or NoSQL databases.	TU
G6.6	Authentication, JWT, state management.	TU
G7.1	Build a frontend application using ReactJS.	TU
G7.2	Manage state in React with hooks and context.	TU
G7.3	Integrate and validate forms with React.	TU

5. TEACHING PLAN THEORY

ID	Topic	Course outcomes	Teaching/Learning Activities (samples)	Assessments
1	Course introduction. Basic HTML and CSS	G2.1, G2.2, G3.1, G3.2	Lecturing, Group discussion, Demonstration	Submit assignment in Moodle
2	Principles of web design. Responsive design with CSS frameworks.	G2.2, G3.3, G4.1	Lecturing, Group discussion, CP#1	Submit assignment in Moodle
3	HTML5 Semantic Structure. Basic JavaScript + DOM	G5.1, G5.3, G5.5, G4.1, G6.1	Lecturing, Q&A	Submit assignment in Moodle
4	Advanced JavaScript. Concepts of CSR/SSR.	G5.1, G5.4, G5.5, G4.1	Demonstration, discussion, HW#1, CP#3	Submit assignment in Moodle

5	Basic ReactJS: Components, Props	G5.2, G4.1, G6.1, G7.1	Lecturing, Demonstration, Q&A	Submit assignment in Moodle
6	ReactJS: State, useState, useEffect, Hooks	G4.1, G5.3, G7.1, G7.2	Lecturing, lab	Submit assignment in Moodle
7	Advanced ReactJS: useContext, react-hook-form/formik, UI Library	G4.1, G5.3, G7.1-G7.3	Case study, Discussion, lab	Submit assignment in Moodle
8	Routing, API Integration (React Router, Public API)	G4.1, G6.6, G7.1-G7.3	Demo, lab, HW#2, CP#3	Submit assignment in Moodle
9	NodeJS Express: Building RESTful API	G6.1, G6.2, G6.3, G6.4	Lecturing	Submit assignment in Moodle
10	CRUD, Middleware, Multer Uploads	G6.4, G4.1	Lab REST, file upload	Submit assignment in Moodle
11	JWT Authentication, State management	G6.6, G4.1, G7.2, G7.3	Lab, Auth: JWT, session vs token, HW#3, CP#4	Submit assignment in Moodle
12	Final Project Presentations	G1.1, G4.2	Presentations	Oral assessment

6. LABORATORY

ID	Topic	Course outcomes	Teaching/Learning Activities (samples)	Assessments
1	Principles of web design. Responsive design with CSS framework	G2.2, G3.3, G4.1	Discuss and answer questions on the course	Submit assignment in Moodle

2	HTML5 Semantic Structure. Basic JavaScript + DOM	G5.1, G5.3, G5.5, G4.1, G6.1	Discuss and answer questions on the course	Submit assignment in Moodle
3	Advanced JavaScript. Concepts of CSR and SSR.	G5.1, G5.4, G5.5, G4.1	Discuss and answer questions on the course	Submit assignment in Moodle
4	Basic ReactJS: Components, Props	G5.2, G4.1, G6.1, G7.1	Discuss and answer questions on the course	Submit assignment in Moodle
5	ReactJS: State, useState, useEffect, Hooks	G4.1, G5.3, G7.1, G7.2	Discuss and answer questions on the course	Submit assignment in Moodle
6	Advanced ReactJS: useContext, react-hook-form/formik, UI Library	G4.1, G5.3, G7.1-G7.3	Discuss and answer questions on the course	Submit assignment in Moodle
7	Routing, API Integration (React Router, Public API)	G4.1, G6.6, G7.1-G7.3	Discuss and answer questions on the course	Submit assignment in Moodle
8	NodeJS Express: Building RESTful API	G6.1, G6.2, G6.3, G6.4	Discuss and answer questions on the course	Submit assignment in Moodle
9	CRUD, Middleware, Multer Uploads	G6.4, G4.1	Discuss and answer questions on the course	Submit assignment in Moodle
10	JWT Authentication, State management	G6.6, G4.1, G7.2, G7.3	Discuss and answer questions on the course	Submit assignment in Moodle

7. ASSESSMENTS

ID	Topic	Description	Course outcomes	Ratio (%)
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A	In-class Assignments			5%
A#1	Reading comprehension, SSR/CSR	Read and present a chosen topic in groups. Understand and compare SSR and CSR	G2.1, G2.2, G3.1, G3.2, G3.3	5%
HW	Homework			55%
HW#1	Weekly assignment	Week topic	G2.1, G5.1, G5.2, G6.1, G7.1–G7.3	5%
HW#2	Dynamic HTML basic	Plain HTML/CSS/JS	G4.1, G5.3-G5.5	15%
HW#3	Dynamic HTML with ReacJS	Using ReactJS with CSS framework	G4.1, G5.4, G7.2	15%
HW#4	RESTful API	Using NodeJS with database, ReacJS and JWT	G4.1, G5.4, G6.4- G6.6, G7.2, G7.3	20%
CP	Course Project			40%
CP#1	Part 1 – Draft idea	Simple Wireframes and basic UI design	G3.4	2.5%
CP#2	Part 2 – Prototype	Responsive static web page	G3.4	2.5%
CP#3	Part 3 – Front-end	React	G5.3, G5.4, G5.5, G7.1–G7.3	5%

CP#4	Part 4 – Back-end	NodeJS + Express: CRUD, JWT auth, Multer upload, DB integration	G6.2, G6.4, G6.5, G6.6	5%
CP#5	Part 5 – Final Oral Assessment	Present full-stack application with demonstration and Q&A	G1.1, G4.2, G6.6, G7.1–G7.3	25%

8. RESOURCES

Textbooks

- Jennifer Niederst Robbins, Learning Web Design (4th edition, O'Reilly)
- Manuel Kiessling, The Node Beginner Book (Leanpub)
- Mike Cantelon et al, Node.js in action (Manning)
- Sandro Pasquali, Mastering Node.js (Packt Publishing)

Tools: Visual Studio Code, PostgreSQL, DBeaver, Node.js, NPM, Git, Web Browsers

Others

- www.w3schools.com

9. GENERAL REGULATIONS & POLICIES

- All students are responsible for reading and following strictly the regulations and policies of the school and university.
- Students who are absent for more than 3 theory sessions are not allowed to take the exams.
- For any kind of cheating and plagiarism, students will be graded 0 for the course. The incident is then submitted to the school and university for further review.
- Students are encouraged to form study groups to discuss the topics. However, individual work must be done and submitted on your own.